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$$\log_2 x = \log_x 2$$

$$\Leftrightarrow \log_2 x = \frac{\log_2 2}{\log_2 x}$$

$$\Leftrightarrow \log_2 x = \frac{1}{\log_2 x}$$

$$\Leftrightarrow (\log_2 x)^2 = 1$$

$$\Leftrightarrow \log_2 x = \pm 1$$

$$\Rightarrow x_1 = 2$$

$$\Rightarrow x_2 = \frac{1}{2}$$

$$\Rightarrow x \in \left\{ \frac{1}{2}, 2 \right\}$$

References